

Lecture 5
–
Outliers

Rotkehlchen / European Robin
(*Erithacus rubecula*)

- It is about 12.5–14.0 cm in length, the male and female are similar in colouration, with an orange breast and face lined with grey, brown upper-parts and a whitish belly.
- The distinctive orange breast of both sexes contributed to the European robin's original name of "redbreast", orange as a colour name being unknown in English until the 16th century, by which time the fruit had been introduced.
- The robin produces a fluting, warbling song during the breeding season with 275 different melodies. Both the male and female sing during the winter, when they hold separate territories.
- The avian magnetic compass of the robin has been extensively researched and uses vision-based magnetoreception, in which the robin's ability to sense the magnetic field of the earth for navigation is affected by the light entering the bird's eye. The physical mechanism of the robin's magnetic sense involves quantum entanglement of electron spins in cryptochrome in the bird's eyes.

Sources:
* Photo by Andhoj on pixabay
* https://en.wikipedia.org/wiki/European_robin

Focusing Objectives

By the end of today's lecture, you will be able to:

1. Understand different causes and impacts of outliers
2. Detect outliers using statistical and visual methods
3. Apply various strategies to handle outliers
4. Decide when to remove or keep outliers based on data context



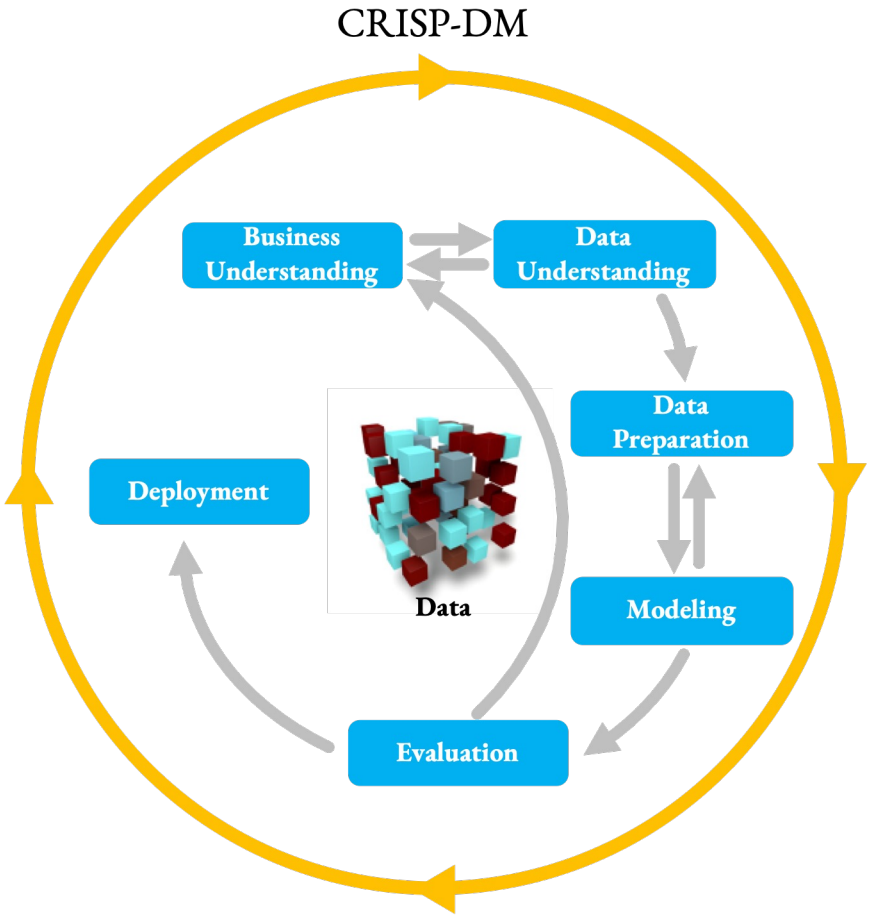
Outliers

1. CRISP-DM: Data Understanding and Preparation – Outliers
2. Reasons for and Consequences of Outliers
3. Types of Outliers
4. Detecting Outliers
5. Handling Outliers

Data Understanding and Preparation Phases of CRISP-DM

Data Understanding & Preparation

- Main goal: gain a comprehensive understanding of the data that will be used for the data mining project and prepare the data for modeling by cleaning, transforming, and selecting the appropriate variables.
- Data quality is being analyzed and improved (using data cleaning methods)
- Additional attributed are created as needed.



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What is an Outlier and Why does it Matter?

What is an Outlier?

- An outlier is a data point that deviates significantly from the rest of the dataset

Consequences of Outliers

- Impact on Machine Learning Models
  - Outliers can skew models, leading to poor predictions
  - Some models (like linear regression and k-NN) are sensitive to extreme values
- Impact on Data Analysis
  - Can distort summary statistics like mean and standard deviation
  - May indicate errors in data collection or highlight meaningful anomalies

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Causes of Outliers

- Data Entry Errors
  - Example: Typos or incorrect values (e.g., entering a salary as $1,000,000 instead of $100,000)
- Measurement Errors
  - Example: Sensor malfunctions, incorrect readings (e.g., a temperature sensor records -300°C)
- Natural Variations
  - Example: Genuine extreme values in populations (e.g., NBA player heights)
- Fraudulent or Anomalous Behaviour
  - Example: Unusual credit card transactions indicating fraud

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Outlier Examples in the Real World

- Finance
  - Stock market crashes, unusually high or low trades
- Healthcare
  - Extremely high or low blood pressure readings
- E-commerce
  - Large purchase amounts that may indicate fraud
- Sports
  - Marathon finishing times with unusually fast or slow runners

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Outliers vs. Noise

	Outliers	Noise
Definition	**Extreme values that deviate significantly from the majority of data**	Random, meaningless fluctuations in data, often from measurement errors or background interference
Cause	Can be real phenomena, errors, or anomalies	Usually caused by random errors, sensor malfunctions, or environmental factors
Impact	May indicate important events, fraud, or true rare occurrences	Generally, does not provide useful information and should be minimized
Detection	Statistical methods (Z-score, IQR), visualization (boxplots, histograms)	Often identified through smoothing techniques or signal processing
Handling	May be removed, transformed, or kept depending on context	Typically filtered out or smoothed using moving averages, signal filtering, etc
Example	An extremely high sales transaction indicating fraud	Random static in an audio recording

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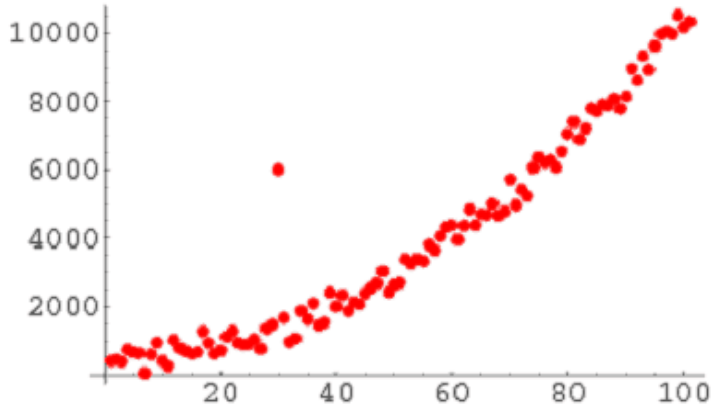
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Types of Outliers 1: Global Outliers (Point Outliers)

- Global Outlier: a single data point that is significantly different from the rest of the dataset
- Examples
  - A house priced at $10M in a neighborhood where most houses cost $200K–$500K
  - A student scoring 100% in a test where most students score 60–80%
  - An intrusion into a computer network

  - never before seen scams/hacking
- Detection Methods
  - Z-score method
  - IQR method
  - Boxplots, histograms

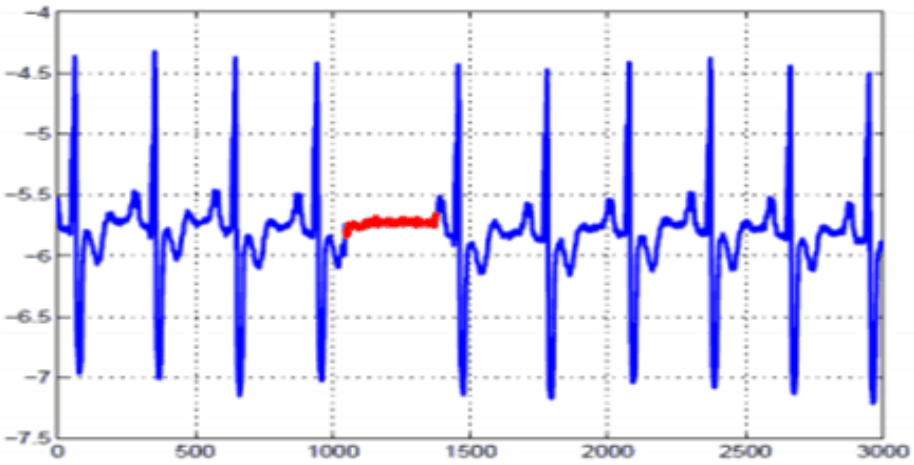


e.g. prediction for house pricing > outliers (especially high prices) would be deleted and range of data boxed in, as a result, outliers would be ignored in both the training and application (e.g. someone is selling a mansion)

Source: https://www.anblicks.com/blog/an-introduction-to-outliers/)

Types of Outliers 2: Collective Outliers

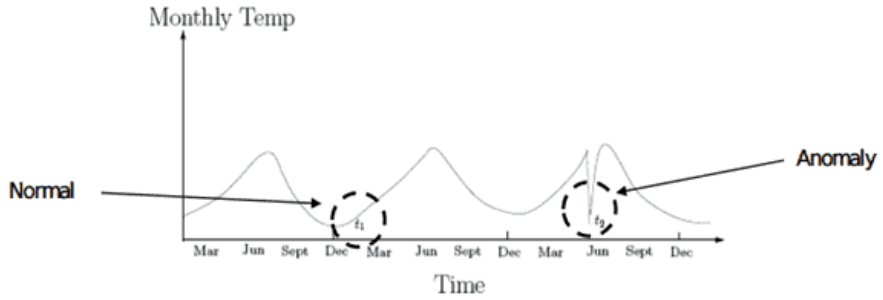
- Collective Outliers: A group of data points that deviate significantly from the expected pattern
- Examples
  - Cybersecurity: A sudden spike in network traffic could indicate a cyberattack
  - Manufacturing: a machine producing a batch of defective products
- Detection Methods
  - Clustering algorithms
    e.g., DBSCAN, k-Means
  - Anomaly detection techniques
    e.g., LOF, Isolation Forest



Source: https://www.anblicks.com/blog/an-introduction-to-outliers/)

Types of Outliers 2: Contextual (Conditional) Outlier

- Contextual Outlier: a data point that is anomalous in a specific context (but not otherwise)
- Attributes of data points should be divided into two groups
  - Contextual attributes: defines the context, e.g., time & location
  - Behavioural attributes: characteristics of the object, used in outlier evaluation, e.g., temperature
- Example
  - Weather: 30°C is normal in summer but an outlier in winter
  - Stock Market: A 5% price drop might be normal on a volatile stock but an outlier on a stable DAX stock
- Detection Methods:
  - Domain knowledge
  - Context-aware statistical models



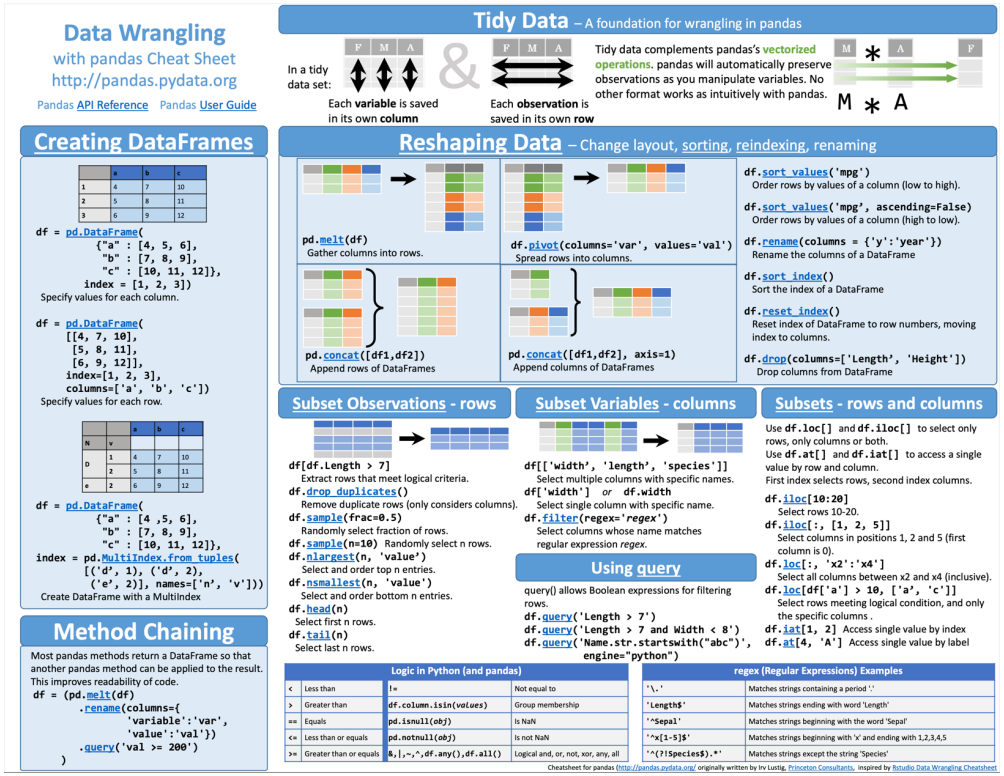
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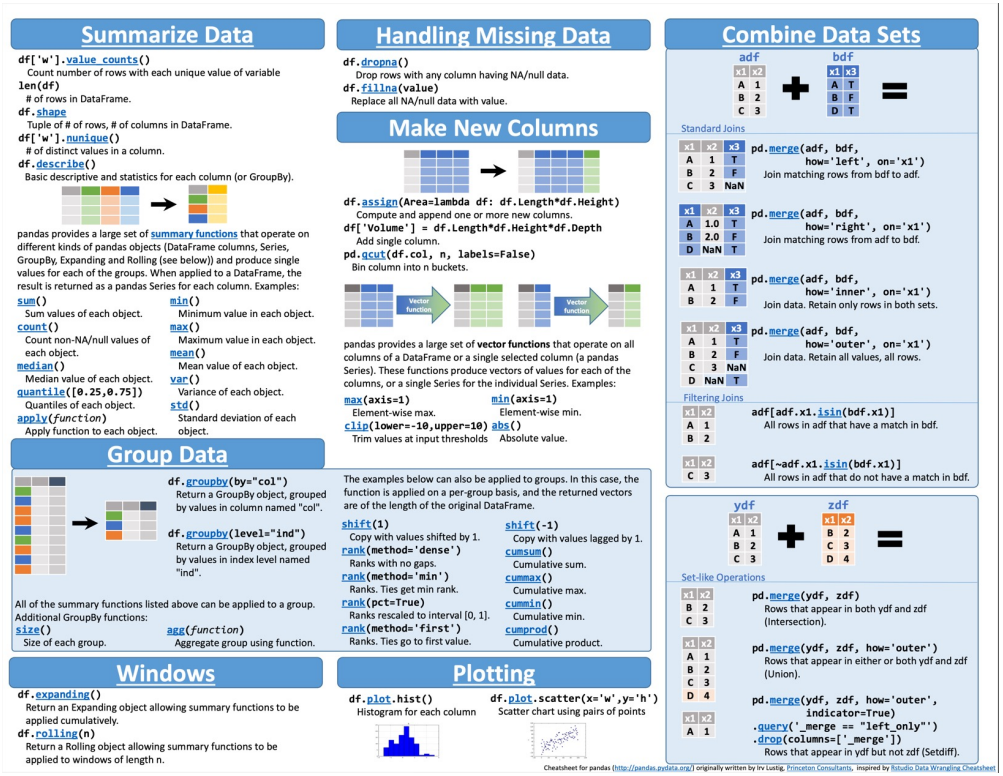
Type of Outliers - Summary

- Outliers can be categorized as
  - Global (Point) Outliers → Individual extreme values
  - Collective Outliers → Anomalous patterns in groups of data points
  - Contextual (Conditional) Outliers → Unusual only in specific contexts
- Each type requires different detection techniques
- Understanding the nature of outliers is key to deciding how to detect and handle them
- Not all outliers are bad, and some may contain valuable insights

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Pandas Cheat Sheet: Helpful for the following Exercises





Exercise

Exercise 1

Iris Data – Loading, Munging, Missing Values

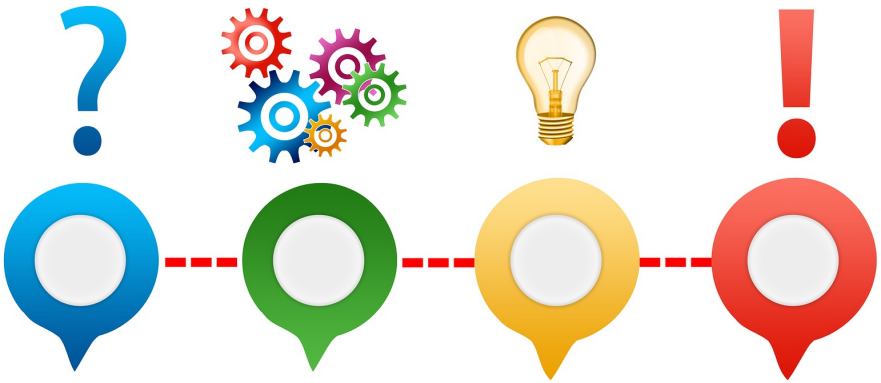


Photo by Gerd Altmann on Pixabay

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Identifying (Global) Outliers with Summary Statistics

- Summary statistics help spot outliers in numerical data
- Outliers often appear in extreme min/max values
- Using describe() in Pandas
- Examples
  - `df.describe()`
  - `df.describe().T`

Identifying (Global) Outliers mathematically using the Z-score

- Z-Score — should be changed according to data, could be lower or higher and depends on the data being in a normal distribution
  - Definition of Outliers: **points that are more than 3 times the standard deviation from the mean**
    - 99.7% of the data are inside 3-times the standard deviation for the normal distribution
  - Computed directly or via z-score scaling
  - Works well for Gaussian and symmetric distributions
  - Example

```
df['Z-score'] = (df['Price'] - df['Price'].mean()) / df['Price'].std()

outliers = df[df['Z-score'].abs() > 3]
```


Source: https://towardsdatascience.com/ways-to-detect-and-remove-the-outliers-404d16608dba

Identifying (Global) Outliers mathematically using the IQR-Score

- IQR-Score
  - Q1 = 25%-percentile (=lower quartile) and Q3 = 75%-percentile (=upper quartile)
  - IQR (interquartile range) = distance between Q1 and Q3
  - Definition of Outliers: **points that are more than 1.5*IQR below Q1 or 1.5*IQR above Q3**

is essentially 3*IQR deviation as previous example, should also be changed according to data

  - Box Plot = visual way to show the IQR values
  - Works well for skewed distributions

  - Example

```
Q1 = df['Price'].quantile(0.25)
Q3 = df['Price'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = df[(df['Price'] < lower_bound) | (df['Price'] > upper_bound)]
```


Source: https://towardsdatascience.com/ways-to-detect-and-remove-the-outliers-404d16608dba

meaning only in one axis, multidimensional outliers cannot be filtered out with these methods

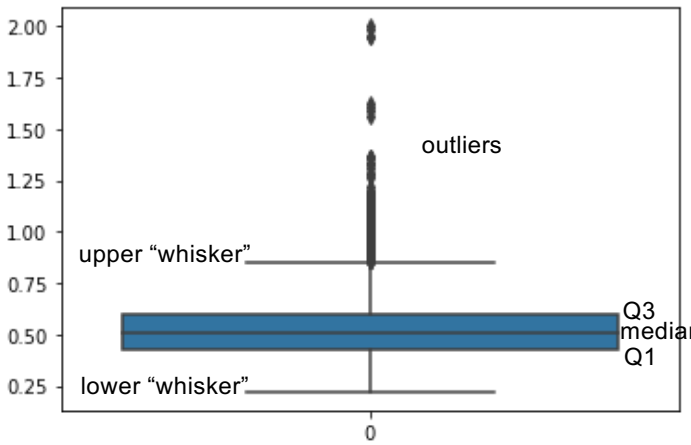
Identifying (Global) univariate Outliers visually using Boxplots

Boxplots: Show the univariate distribution of data and highlight outliers as individual points

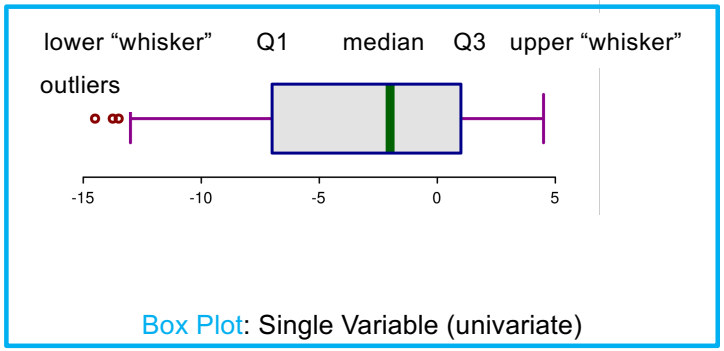
- Outliers appear outside the whiskers (1.5 * IQR range)
- Example
  `df.boxplot(column=['Price'])`

  common method, so pandas has a function for it

- Alternatives
  - Histograms:
    Show data distribution and highlight extreme values
  - Kernel Density Estimation (KDE) Plots:
    Visualize the shape of the distribution
  - Examples:
    - See next week!



Box Plot: Single Variable (univariate)



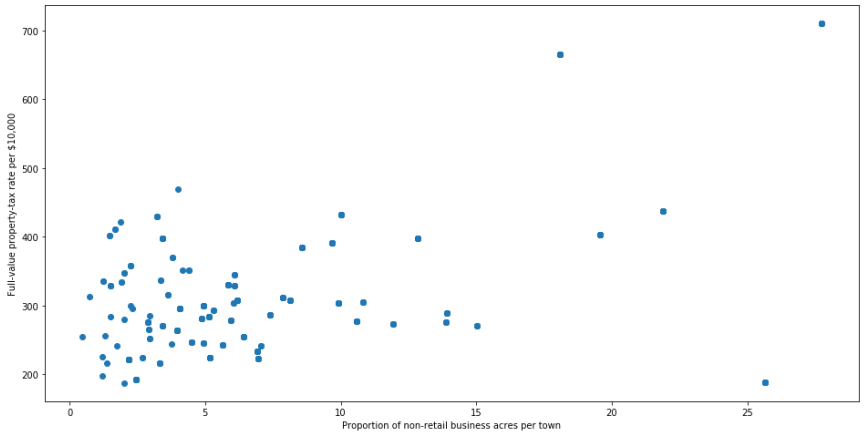
Box Plot: Single Variable (univariate)

Source: https://towardsdatascience.com/ways-to-detect-and-remove-the-outliers-404d16608dba

box is extended until the whiskers hit one of first data points outside of the given range

Identifying (Global) multivariate Outliers visually using Scatter Plots

- Scatter Plots: Help identify outliers in multivariate datasets
  - Useful when outliers affect relationships between variables
- Example
  - See next week!



Scatter Plot: Two Variables (multivariate)

Source: https://towardsdatascience.com/ways-to-detect-and-remove-the-outliers-404d16608dba

Choosing the Right Detection Method for (Global) Outliers

- Boxplot
  - Quick visual detection — Example: Spotting anomalies in income data
- Histogram and KDE Plot
  - Understanding distributions — Example: Checking for fraudulent transactions
- Scatter Plot
  - Multivariate outliers — Example: Identifying unusual sales patterns
- Z-Score
  - Normally distributed data — Example: Detecting salary outliers
- IQR
  - Skewed distributions — Example: Detecting house price outliers

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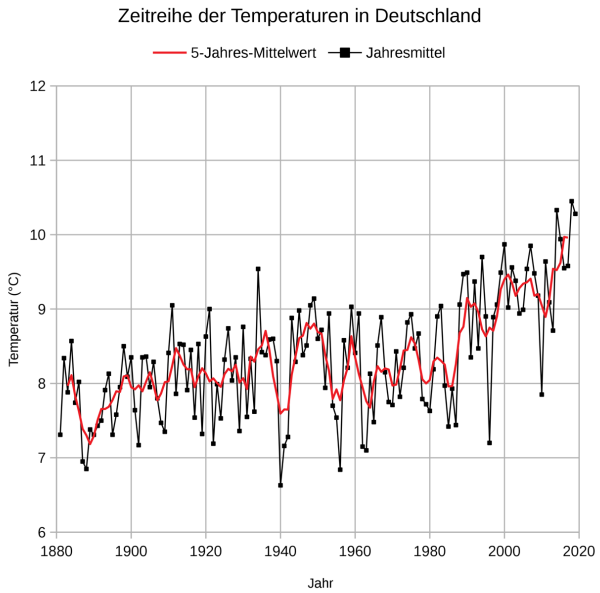
Identifying Collective Outliers

- Much harder than Global Outlier Identification
- Often non-noise, but real “signal” – in this case, opportunity to find something important
- Some Clustering Algorithms can be used
  - e.g., EM-Clustering, DBSCAN
- Specific Outlier/Anomaly Detection Algorithms exist
  - e.g., LOF, Isolation Forests, Autoencoder-based approaches
- Collective Outliers in Time-Series Data
  - Compute rolling mean and standard deviation
  - Value deviating significantly from the rolling mean may be part of an Outlier event

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Identifying Contextual Outliers

- Recap: Attributes of data points should be divided into two groups
  - Contextual attributes: defines the context, e.g., season
  - Behavioural attributes: characteristics of the object, used in outlier evaluation, e.g., temperature
- Compare values within groups
  - Group the data by Contextual attribute categories (such as seasons)
  - Apply outlier detection methods (e.g., Boxplots or Z-Score) on Behavioural attributes within each group
- Time Series Data
  - Model both seasonality and trend (manually or automatically) and identify Deviations



Source: https://de.wikipedia.org/wiki/Folgen_der_globalen_Erw%C3%A4rmung_in_Deutschland

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Strategies for Handling Outliers

- Remove Outliers: Delete extreme values if they are errors
- Cap Outliers (Truncation, Winsorization): Limit extreme values to a predefined range
  could lead to a small spike at range bounds
- Transform Outliers: Apply log or power transformations
- Imputation: Replace extreme values with statistical estimates
- Use Robust Models: Some ML algorithms handle outliers better than others
- Dual Analysis: Run analysis with and without Outlier
  do modeling with and without outliers and compare (possibly expensive)

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Removing or Capping Outliers

- Remove Outliers
  - Use Z-score or IQR to identify and remove outliers
  - Risk: Removing too many values can reduce data quality
  - Example:

```
df['Z-score'] = (df['Price'] - df['Price'].mean()) / df['Price'].std()
df_filtered = df[df['Z-score'].abs() <= 3]
```

- Cap Outliers (Truncation, Winsorization)
  - Capping replaces extreme values with threshold values
  - Good for: Keeping data size intact while reducing outlier impact
  - Example using the 5th and 95th percentiles:

```
cap_lower = df['Price'].quantile(0.05)
cap_upper = df['Price'].quantile(0.95)

df['Price_Capped'] = np.where(df['Price'] > cap_upper, cap_upper,
        np.where(df['Price'] < cap_lower, cap_lower, df['Price']))
```


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Transforming Outliers

- Applying Log Transformations
  - Log transformation reduces the impact of outliers
  - Useful for highly skewed distributions
  - Example:
    ```
    df['Price_Log'] = np.log(df['Price'])
    ```

- Applying Power Transformations (Box-Cox & Yeo-Johnson)
  - Box-Cox Transform (requires positive values).
  - Yeo-Johnson Transform (handles zero and negative values)
  - Useful when applying machine learning models that assume normal distributions
  - Example:
    ```
    from scipy import stats
    df['Price_BoxCox'], _ = stats.boxcox(df['Price'])
    ```


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Imputation or Using Robust Models

- Imputation
  - Useful for small datasets where removing data could significantly impact analysis
  - See lecture on Missing Values

- Using Robust Models
  - Some ML models are resistant to outliers
  - Examples
    - Tree-based models (e.g., Decision Trees, Random Forests, Extreme Gradient Boosting)
    - Robust regression (e.g., Huber Regression, RANSAC)

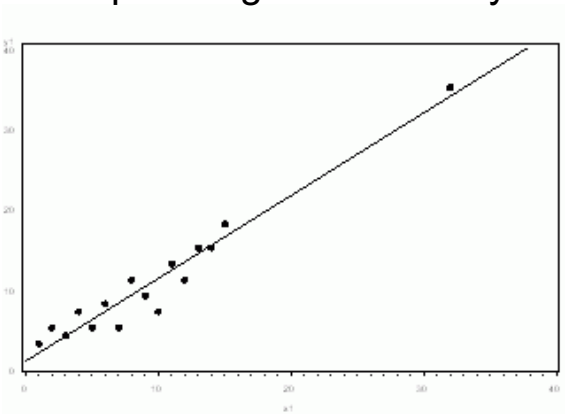
  - Some ML models are very sensitive to outliers
  - Examples
    - Linear Regression, k-NN, PCA

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Dual Analysis

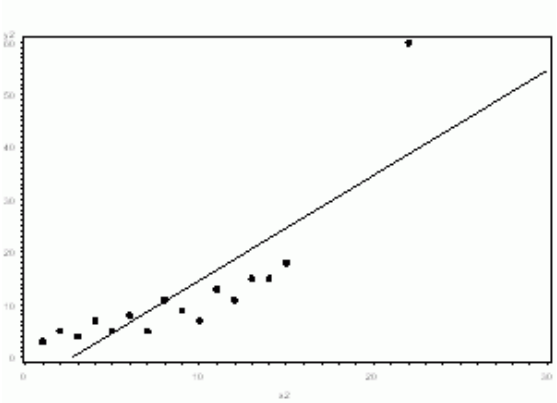
- Dual Analysis: Run analysis with and without Outlier
  - If it does not change the result and assumptions: may remove Outlier and note this
  - If it does change the result, but not the assumptions: show both results and discuss changes
  - It if creates results: drop, note this, and show results without Outlier (and investigate Outlier further)
  - Useful if we are unsure if the Outlier is Noise or not

  - Example: Regression Analysis with an Outlier for which we are unsure if it is Noise

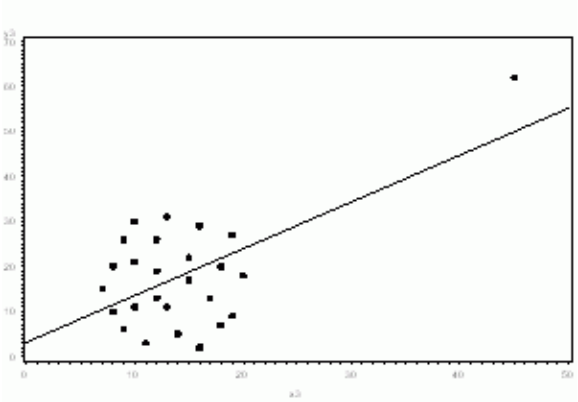


(a) Outlier does not change the result and assumptions

(e.g. that data has a linear upward trend)



(b) Outlier does change the result, but not the assumptions
> worsens model by skewing result



(c) Outlier creates the result

> "improves" the model by determining the model

Source: https://www.theanalysisfactor.com/outliers-to-drop-or-not-to-drop/

not detectable with train-test split
cross-validation would detect it

outlier could also be in target
constraining can be done both on input and output

Handling Outliers in Practise

- If the Outlier is Noise/an Error: Remove Outlier
- If the Outlier is not Noise, or if we are unsure
  - *Best method depends on the dataset and business context*
  - Cap
    - If extreme values distort data. Risk: Can hide valuable trends
  - Transform
    - If data is skewed. Risk: Hard to interpret data
  - Impute
    - If data size is small. Risk: May introduce bias
  - Dual Analysis
    - If we have the time/resources and an interpret the model. Risk: time/resource intensive

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Exercise

Exercise 2

Iris Data – Outlier Detection & Handling, Merging

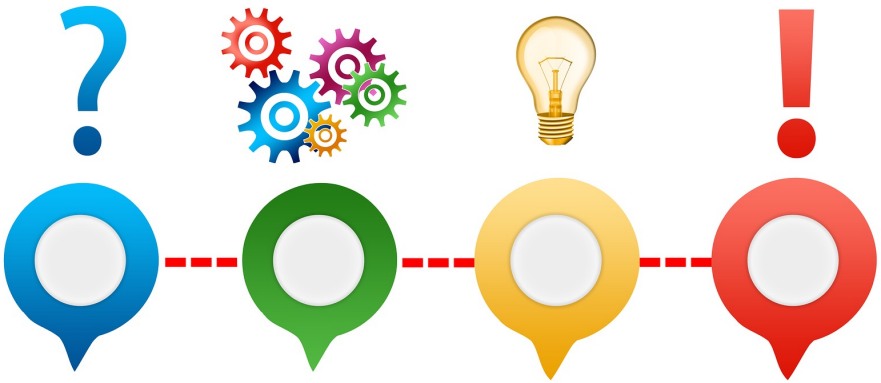
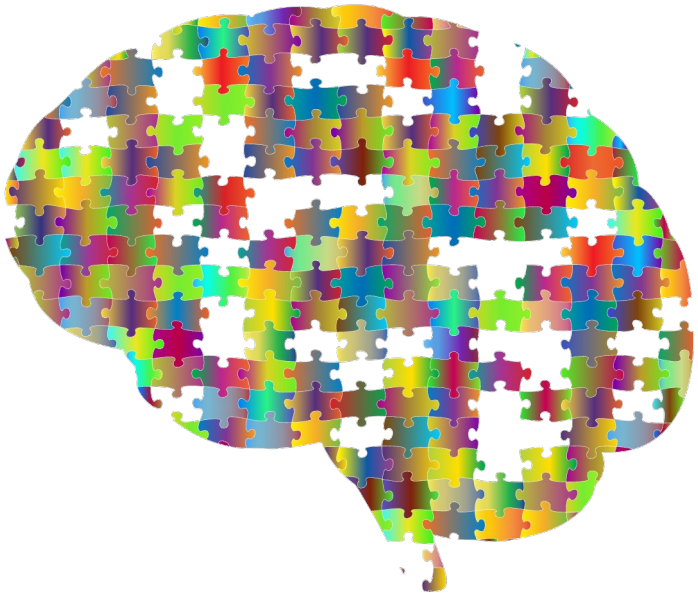
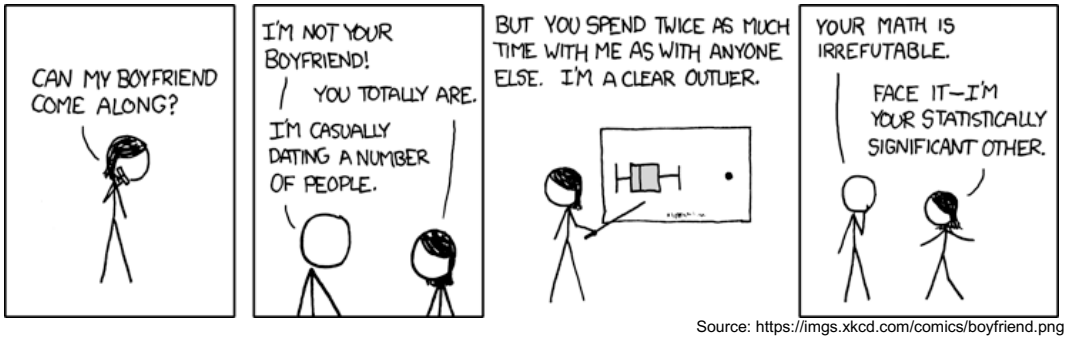


Photo by Gerd Altmann on Pixabay

Key Takeaways



- Outlier = data point that deviates significantly from the rest of the dataset
- Cause: Error, Real Phenomena (Distribution), Anomalies
- Types of Outliers – Global, Collective, Contextual Outliers
  - Noise and Outliers
- Detecting Outliers: depends on Type of Outlier
  - Visually (box plots, scatter plots)
  - Mathematically (z-score, IQR score)
- Handling Outliers
  - Remove Outliers
  - Cap Outliers
  - Transform Outliers
  - Imputation
  - Use Robust Models
  - Dual Analysis



Source: https://imgs.xkcd.com/comics/boyfriend.png

Image by Gordon Johnson on pixabay